

COMMUNICATION & NETWORK FUNDAMENTALS

COURSE PROJECT REPORT

MEMBERS:

	Name	Id
1	Alhawra Ali Alhamel	2240002231
2	Zainab Jamal Altarouti	2240005573
3	Dhai Ali Najdi	2240006429
4	Faridah Hussain Abdulaziz	2240002838
5	Zainab Zaki Almajdi	2240005137

Instructor :Dr. Fatimah M Alhassan

Introduction :

We will go over our topology's steps and the project's final version in this report. To complete the project, model the design, and set up the network connection, we used Cisco Packet Tracer. We selected template 3 because we thought it was well-structured and simple to design.

Table of Addresses:-

Device	Interface	IP Address	Subnet Mask	Default Gateway
Router 0	Fa 0/0	192.168.1.0	255.255.255.0	N/A
	Se 0/1/1	10.0.0.0	255.0.0.0	N/A
	Se 0/1/0	11.0.0.1	255.0.0.0	N/A
Router 1	Fa 0/0	192.168.2.1	255.255.255.0	N/A
	Fa 0/1	192.168.0.1	255.255.255.0	N/A
	Se 0/1/0	11.0.0.2	255.0.0.0	N/A
	Se 0/1/1	12.0.0.2	255.0.0.0	N/A
Router 2	Fa 0/0	192.168.3.1	255.255.255.0	N/A
	Fa 0/1	192.168.4.1	255.255.255.0	N/A
	Se 0/1/1	12.0.0.1	255.0.0.0	N/A
Router 3	Fa 0/0.10	192.168.10.1	255.255.255.0	N/A
	Fa 0/0.20	192.168.20.1	255.255.255.0	N/A
	Se 0/0/0	10.0.1	255.0.0.0	N/A

Switch 0	Fa 0/1 – Fa 0/3	N/A	N/A	N/A
Switch 1	Fa 0/1 – Fa 0/3	N/A	N/A	N/A
Switch 2	Fa 0/1 – Fa 0/3	N/A	N/A	N/A
Switch 3	Fa 0/1 – Fa 0/6	N/A	N/A	N/A
Switch 4	Fa 0/1 – Fa 0/3	N/A	N/A	N/A
Switch 5	Vlan 10	N/A	N/A	N/A
	Vlan 20			
Switch 6	Fa 0/1 – Fa 0/4	N/A	N/A	N/A
PC 0	NIC	192.168.1.2	255.255.255.0	192.168.1.1
PC 1	NIC	192.168.1.3	255.255.255.0	192.168.1.1
PC 2	NIC	192.168.2.2	255.255.255.0	192.168.2.1
PC 3	NIC	192.168.2.3	255.255.255.0	192.168.2.1
PC 4	NIC	192.168.3.2	255.255.255.0	192.168.3.1
PC 5	NIC	192.168.3.3	255.255.255.0	192.168.3.1
PC 6	Wireless	192.168.4.7(DHCP)	255.255.255.0	192.168.4.1
PC 7	NIC	192.168.0.2	255.255.255.0	192.168.2.1
PC 8	NIC	192.168.0.3	255.255.255.0	192.168.2.1
PC 9	NIC	192.168.20.11(DHCP)	255.255.255.0	192.168.20.1
PC 10	NIC	192.168.20.12(DHCP)	255.255.255.0	192.168.20.1
PC 12	Wireless	192.168.4.10(DHCP)	255.255.255.0	192.168.4.1

Laptop 0	NIC	192.168.4.4(DHCP)	255.255.255.0	192.168.4.1
Laptop 1	NIC	192.168.4.3(DHCP)	255.255.255.0	192.168.4.1
Laptop 2	Wireless	192.168.4.6(DHCP)	255.255.255.0	192.168.4.1
Laptop 4	Wireless	192.168.4.9(DHCP)	255.255.255.0	192.168.4.1
Printer 0	FastEthernet	192.168.4.5(DHCP)	255.255.255.0	192.168.4.1
Printer 1	Wireless	192.168.4.8(DHCP)	255.255.255.0	192.168.4.2
Printer 2	Wireless	192.168.4.6(DHCP)	255.255.255.0	192.168.4.1
Server	FastEthernet	192.168.4.2	255.255.255.0	192.168.4.1
IP Phone 0	FastEthernet	192.168.10.2	255.255.255.0	192.168.10.1
		Line Number: 123		
IP Phone 1	FastEthernet	192.168.10.3	255.255.255.0	192.168.10.1
		Line Number: 456		

Topology:-

Our design consists of:

1.Routers:

Router 3, Router 1, and Router 2 connect different subnets and provide routing between them and the external network

2.Switches:

Manage local traffic within each subnet, connecting devices like PCs, IP phones, printers, and laptops

3.Devices

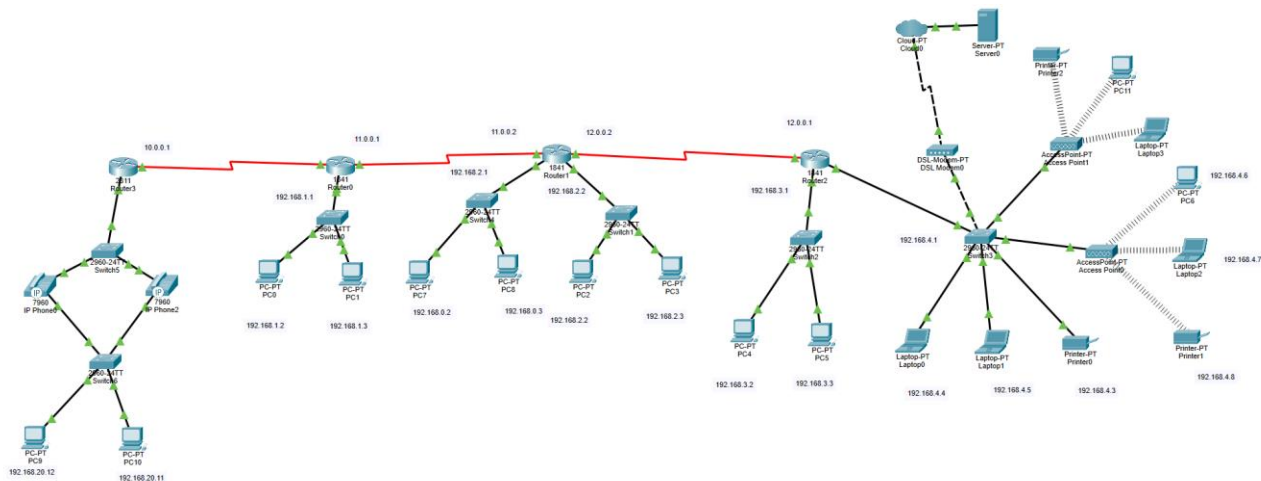
- PCs and IP phones are connected to switches in each subnet for internal communication.
- Printers are networked for shared access within their subnets.

4.Laptops and Access Points:

Laptops connect to switches or access points, enabling wireless network access

5.Servers and Modem

A Server provides centralized services connected to the cloud and connected to the modem

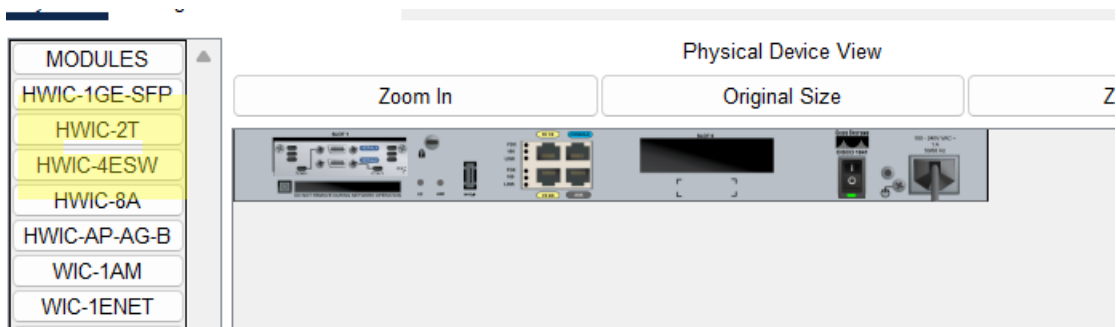


[Packet tracer file](#)

[Video simulation](#)

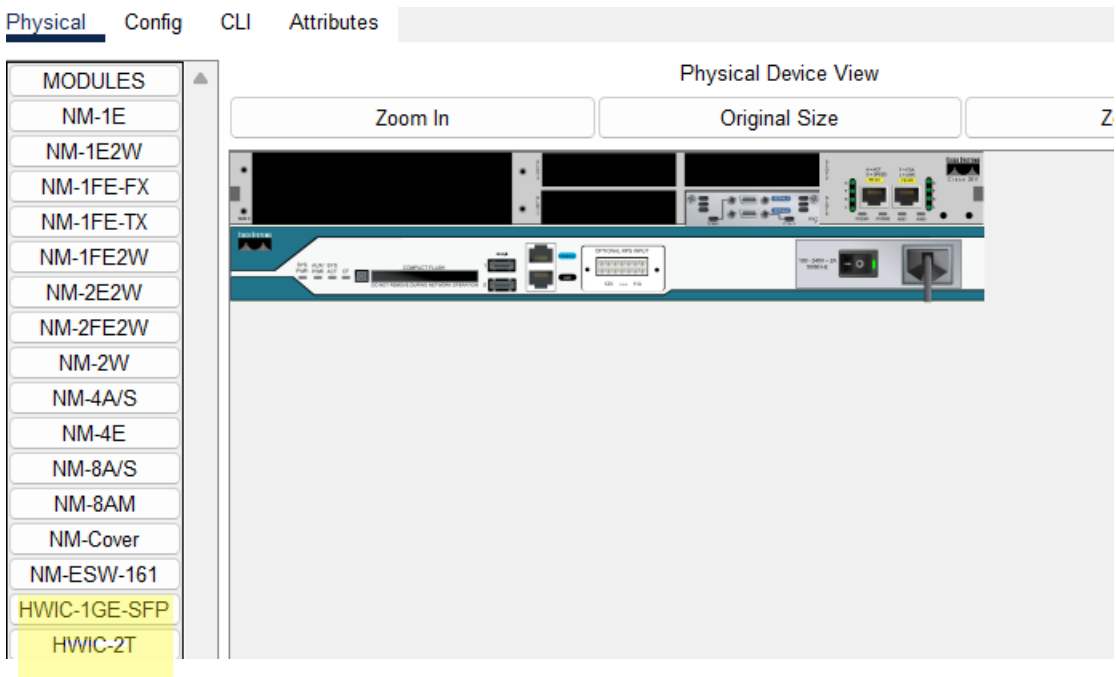
Step 1:

We connected the module (HWIC-2T) to the routers, to be able to connect the routers via Serial DCE cable.



Router 1841:

Router 2811:



Step 2:

We start configuring each IP address in the network (router, switch, devices) with the correct gateway of the router.

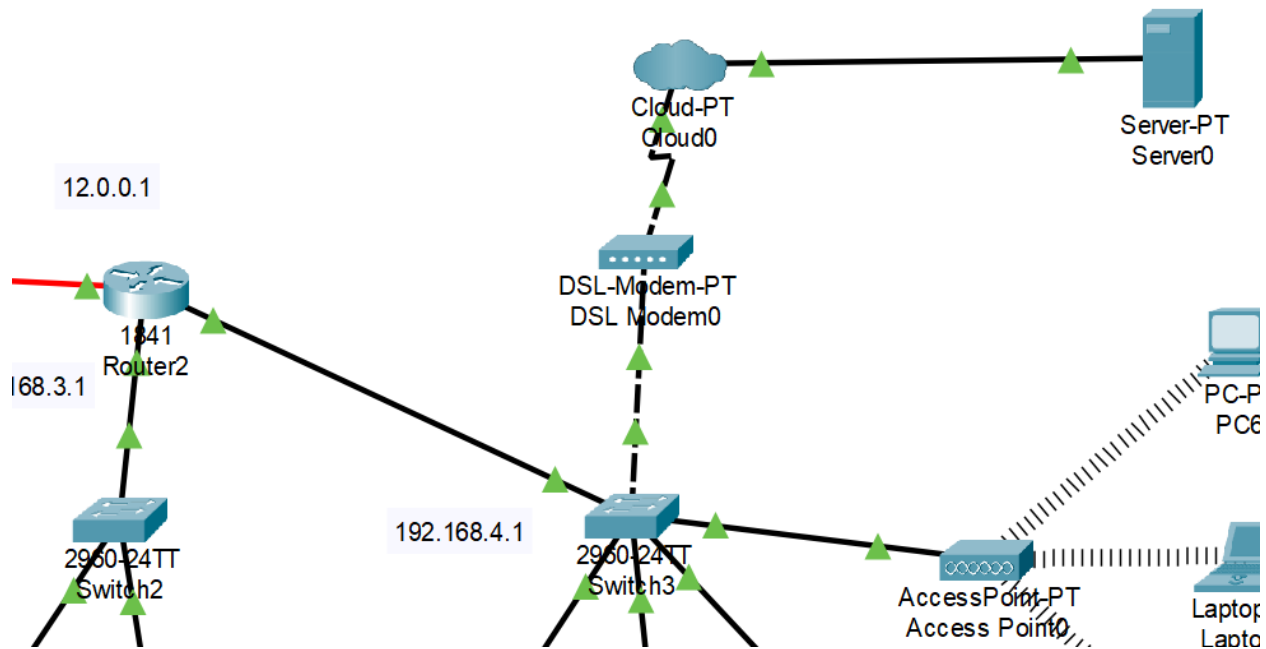
Step 3:

Routing routers so it could identify other networks and send and receive from different networks through gateways and network IP addresses.

Network Address
192.168.2.0/24 via 11.0.0.2
192.168.3.0/24 via 11.0.0.2
12.0.0.0/8 via 11.0.0.2
192.168.4.0/24 via 11.0.0.2
192.168.0.0/24 via 11.0.0.2
192.168.20.0/24 via 10.0.0.1

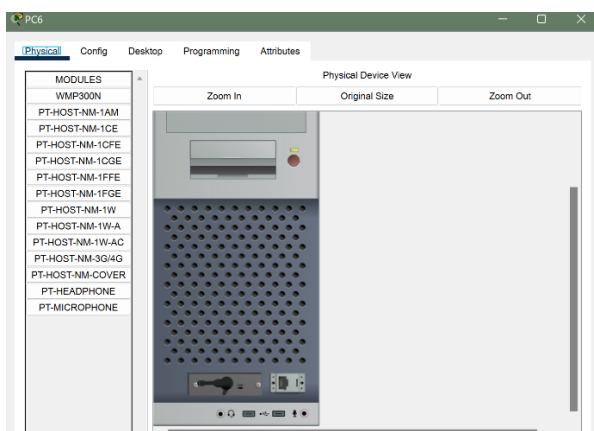
Step 4:

Here we added the cloud, another network, and a server. The server, using DHCP automatically assigns IP addresses to the devices in the network to make things easier. **We also added the new network to the router so it could communicate with other networks.**



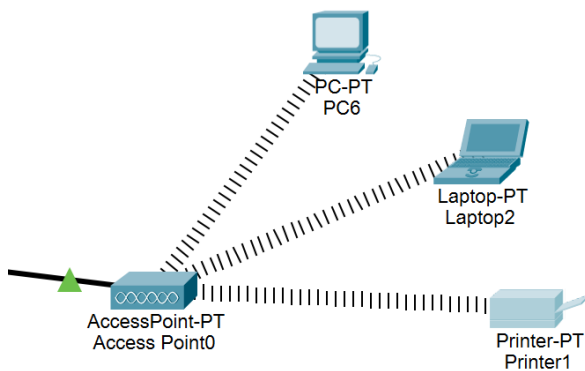
Step 5:

We added an access point because it is necessary to establish a wireless connection. Devices can connect to the access point using its assigned name and password.



Step 6:

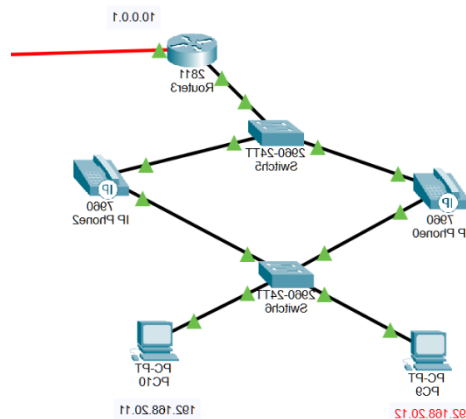
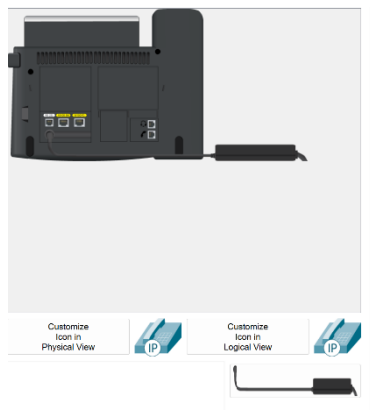
Connecting devices to a wireless network requires a module that enables them to connect wirelessly. We connect to the access point using the name and **password** that were assigned earlier.



Step 7:

To add a phone to the network, we

should



first connect the phone to power. Then, connect Phone0 and Phone2 to a switch

that is connected to the router. After that, connect them to another switch, where we will connect PC9 and PC10 in the same switch.

Step 8: We configured two interfaces with names 'Data' and 'Voice.' Then, we set Fa0/1 to trunk mode.

```
Switch(config-if)#
Switch(config-if)#vlan 10
Switch(config-vlan)#name Voice
Switch(config-vlan)#vlan 20
Switch(config-vlan)#name Data
Switch(config-vlan)#exit
Switch(config)#int range fa0/1-15
Switch(config-if-range)#switchp
Switch(config-if-range)#switchport m
Switch(config-if-range)#switchport mode acc
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#sw
Switch(config-if-range)#switchport ac
Switch(config-if-range)#switchport access vlan 20
Switch(config-if-range)#swi
Switch(config-if-range)#switchport Voice vlan 10
Switch(config-if-range)#exit
Switch(config)#int fa0/1
Switch(config-if)#sw
Switch(config-if)#switchport mo
Switch(config-if)#switchport mode t
Switch(config-if)#switchport mode trunk
```

Conclusion:

By connecting 12 PCs, 4 laptops, 3 printers, 4 routers, 7 switches, a server, and 2 access points, all devices can successfully communicate with every other device. We concluded that the network accomplishes every objective listed in the instruction manual. By examining the link between any two locations in the topology to guarantee the intended outcome, we can assess the network's efficacy.

